9.3.4 Eutrophication

The addition of excess quantities of nutrients to aquatic ecosystems termed eutrophication often cause damage on the communities involved. Eutrophication is an enrichment of water by nutrients, especially nitrogen and phosphorus which are maintained at high levels that causes structural changes to the eco system. The sudden influx of abundant nutrients encourages a heavy surface growth of cyanobacteria and algae called a bloom (Figure 9.6). This heavy matt of biomass effectively shuts off the oxygen supply to the lakes below. The lack of oxygen greatly disturbs the ecological balance of the community. It causes massive death of strict aerobes. Fish invertebrates and anaerobic or facultative microbes will survive.

Effects of eutrophication

- Eutrophication can have serious, long term effect. The most notable effect of eutrophication is **algal blooms**.
- When a bloom occurs, the river, lake or ocean becomes covered with algae, which is usually bright green.
- Blue green algae like *Microcystis, Anabaena, Gonyaulax* and other Dinoflagellates produce toxins which are poisonous to aquatic lives such as fish.
- Algal toxin may also contaminate the drinking water supply.
- Plankton blooms of green alga create problems of O_2 supply in the water.
- Musty tastes and odors in drinking water are the other effects of Eutrophication.
- Excessive growth of aquatic weeds which impair fishing, swimming, boating, shell fish production.
- Irrigation canals may become clogged.

Control measures

Five different methods have been suggested to control eutrophication they are,

1. Ecological management (control the flow of nutrients into natural wastes)
2. Advanced treatments Example: Phosphate removal is effected by precipitation with lime.
3. Chemical algicides Example: copper sulphate

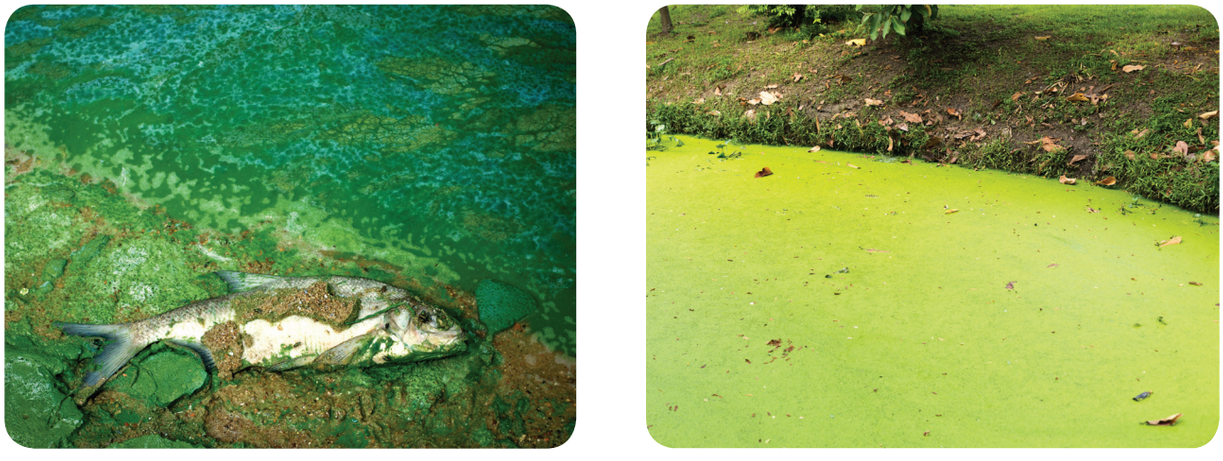


Figure 9.6: Algal blooms in a polluted eutrophic lake

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